

The Effects of Algorithms on Bookings

Evidence from Expedia

Dili Maduabum

April 21, 2026

1 Introduction

Every time a traveler searches for a hotel on Expedia, a ranking algorithm determines which properties appear first. Because attention is highly concentrated at the top of the results page, this ranking can strongly shape which hotels are clicked and booked. Yet whether algorithmic ranking actually improves market outcomes relative to a simpler alternative is ultimately an empirical question.

This paper studies that question using a randomized experiment ran by Expedia (Expedia, 2013). For a subset of search sessions, Expedia displayed hotels in random order rather than in the platform’s algorithmic order. Because this assignment occurred at random before users viewed the results page, it creates credible exogenous variation in exposure to algorithmic ranking.

I use this variation to estimate two causal effects. First, I estimate the intention-to-treat effect of algorithmic ranking on hotel clicks and bookings relative to random ordering. Second, I use ranking assignment as an instrument for hotel position to estimate the causal effect of appearing higher on the page.

The paper contributes reduced-form evidence on the effects of platform ranking algorithms in online travel markets. The closest related work is Ursu (2018), who uses the same data to estimate a structural model of consumer search on Expedia. The main finding is that algorithmic ranking reduces both clicks and bookings relative to random ordering, suggesting that Expedia’s ranking system did not improve average booking outcomes in this setting.

2 Data

Expedia is an online travel platform where users enter a destination, travel dates, and party size, and receive a ranked list of available hotels. The Expedia Personalized Sort dataset (Expedia, 2013) contains logs of these searches, with one row per (search, hotel)

pair. The full training set covers approximately 9.9 million hotel listings across 400,000 unique searches.

The key variable is `random_bool`, which equals one when a search received hotels in random order and zero when the algorithm ordered the results. Expedia assigned this condition at the search level before users saw any results, making it independent of hotel quality, user characteristics, and anything else by design. The two outcomes are `booking_bool` (did the user book this hotel?) and `click_bool` (did the user click on this hotel?). The endogenous variable for the IV analysis is `position`, the hotel’s rank on the results page, where position 1 is the top.

Fourteen pre-specified cleaning rules are applied before analysis. The most important are: dropping rows with missing outcomes, missing prices, or unrated hotels (star rating of zero); imputing missing visitor purchase history to zero and adding an indicator flag; winsorizing `price_usd` at the 99th percentile (\$530) to remove outliers. To keep computation tractable, 50,000 searches are randomly sampled at the search level, preserving within-search structure. The final analytic sample contains 1,221,840 (search, hotel) pairs across 50,000 searches.

Table 1 reports the distribution of searches and raw outcomes by condition. About 94% of searches received algorithmic ranking (47,064 searches; 1,153,157 hotel listings) and 6% received random ordering (2,936 searches; 68,683 listings). The raw booking rate is 4.1% under algorithmic ranking and 4.3% under random ranking. The raw click rate is 4.5% versus 4.7%. These differences foreshadow the main finding.

3 Potential Outcomes and Assumptions

3.1 Potential Outcomes

Let i index (search, hotel) pairs. Define $Z_i = 1$ if search i received algorithmic ranking and $Z_i = 0$ if it received random ranking. For each hotel listing, potential outcomes $Y_i(1)$ and $Y_i(0)$ denote the booking outcome under algorithmic and random ranking respectively. The ITT estimand is the average effect of the algorithm across all (search, hotel) pairs:

$$\tau_{\text{ITT}} = E[Y_i(1) - Y_i(0)]$$

For the IV analysis, let D_i denote the hotel’s displayed position, with $D_i(1)$ and $D_i(0)$ denoting the position the hotel would receive under algorithmic and random ranking respectively. Rather than comparing ranking regimes directly, the IV design asks: what is the causal effect of being shown one position higher on the page? Define $Y_i(d)$ as the booking outcome when hotel i appears at position d . The LATE is identified as the ratio of two reduced-form effects, both of which are estimable from the randomization:

$$\tau_{\text{LATE}} = \frac{E[Y_i(1) - Y_i(0)]}{E[D_i(1) - D_i(0)]} = \frac{\tau_{\text{ITT},Y}}{\tau_{\text{ITT},D}}$$

The numerator $\tau_{ITT,Y}$ is the effect of the instrument on booking. The denominator $\tau_{ITT,D}$ is the first-stage effect of the instrument on position. Dividing one by the other scales the booking effect by how much the instrument actually moves positions, yielding the effect of a one-position change in position for complier hotels: hotels whose displayed position changes when `random_bool` switches from zero to one.

3.2 SUTVA

SUTVA requires that a hotel’s outcome depends only on its own treatment, not on what happens to other hotels in the same search. This is clearly an approximation here. When hotel *A* is moved to position 1, hotel *B* is pushed to position 2. Hotel *B*’s outcome depends directly on what happened to hotel *A*. This within-search interference is inherent to any ranking experiment and means ITT estimates should be interpreted as average effects across the full (search, hotel) sample, not as clean individual-level causal effects. This is partially addressed by defining the unit of randomization as the search session and clustering interpretation accordingly.

A second concern is that “algorithmic ranking” is not a single intervention. The algorithm places different hotels at different positions depending on their characteristics, so the treatment varies across units. This multiple-versions problem motivates the IV design, which asks a cleaner question: holding everything else fixed, what does it cost a hotel to appear one position lower on the page?

3.3 Identification Assumptions

The ITT requires only that `random_bool` is independent of potential outcomes. This holds by Expedia’s randomization design. The IV analysis requires four additional conditions.

Relevance. The instrument must shift hotel positions. This is verified directly: [Table 7](#) shows a strong first stage with F-statistics above 91.

Exclusion restriction. The instrument affects booking only through its effect on position, not through any other channel. The main threat would be users noticing that results look random and searching differently as a result. Since the randomization is invisible in the Expedia interface, this seems unlikely, though it is formally untestable.

Monotonicity. No hotel should appear at a lower position under random ranking than under algorithmic ranking. In [section 6](#) this assumption is examined across hotel quality tiers and found to be questionable.

Positivity. Every hotel has a positive probability of being assigned any position under both conditions, which holds by construction under random ordering.

4 Methods

4.1 ITT: Difference-in-Means and Covariate Adjustment

Because `random_bool` is randomly assigned, the simple difference in mean outcomes between conditions is an unbiased estimator of the ITT:

$$\hat{\tau}_{\text{DiM}} = \bar{Y}_{Z=1} - \bar{Y}_{Z=0}$$

A covariate-adjusted estimate is also reported, which regresses the outcome on the treatment indicator and a vector of pre-treatment hotel and search characteristics:

$$Y_i = \alpha + \tau Z_i + X_i' \beta + \varepsilon_i$$

The coefficient τ on Z_i is the adjusted ITT estimate. The vector X_i includes hotel characteristics (star rating, review score, brand indicator, location quality score, log historical price, promotion flag) and search characteristics (length of stay, booking window, party size, Saturday night indicator, visitor purchase history, destination distance). Including these covariates does not change the identification strategy since Z_i is randomly assigned. Instead it isolates the treatment effect from residual variation attributable to observable hotel and search characteristics, which reduces standard errors and improves precision. All specifications use HC2 heteroskedasticity-robust standard errors. HC2 is preferred over conventional OLS standard errors because the (search, hotel) pair observations are not identically distributed: hotels differ in their variance of outcomes, and HC2 accounts for this without requiring any parametric assumption on the error structure.

4.2 IV: Wald Estimator

To estimate the effect of position directly, `random_bool` is used as an instrument for `position`. The intuition is that searches randomly assigned to `random_bool`=1 scramble hotel positions relative to what the algorithm would have chosen. This generates quasi-random variation in position that identifies how booking probability changes when a hotel moves one slot down the page. Concretely, the Wald estimator plugs the DiM estimates of the two reduced-form effects into the LATE formula:

$$\hat{\tau}_{\text{LATE}} = \frac{\hat{\tau}_{\text{ITT},Y}}{\hat{\tau}_{\text{ITT},D}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

where $\bar{D}_{Z=1} - \bar{D}_{Z=0}$ is the first-stage effect of the instrument on position. A covariate-adjusted IV estimate is also reported using `iv_robust`, adding the same X_i vector to both the first and second stages.

5 Results

5.1 Balance

Figure 1 shows a Love plot of absolute standardized mean differences (SMD) across 15 covariates by condition. Since `random_bool` was randomly assigned, covariates should look similar across conditions. Fourteen of 15 covariates have $|\text{SMD}| < 0.10$, consistent with valid randomization. Two hotel-level covariates, `prop_location_score1` and `promotion_flag`, have $|\text{SMD}|$ marginally above 0.10 (both 0.123). This mild imbalance is expected: randomization was at the search level while the unit of observation is the (search, hotel) pair, and hotels within the same search are correlated. This can induce finite-sample imbalance on hotel-level characteristics even under valid randomization. Both variables are included as controls in the adjusted specifications.

5.2 ITT Estimates

Table 7 and Figure 2 report ITT estimates for both outcomes. For bookings, the unadjusted DiM estimate is -0.0019 (HC2 SE = 0.0008, $p = 0.015$) and the covariate-adjusted estimate is -0.0035 (SE = 0.0010, $p < 0.001$). For clicks, the DiM is -0.0023 (SE = 0.0008, $p = 0.005$) and the adjusted estimate is -0.0042 (SE = 0.0010, $p < 0.001$).

All four estimates are negative and statistically significant. To put these numbers in context: the baseline booking rate is about 4.1% under algorithmic ranking. An ITT of -0.0035 means that switching from random to algorithmic ranking is associated with roughly a 9% relative reduction in booking probability. That is not a small effect for a platform operating at Expedia’s scale. The effect on clicks is slightly larger than on bookings, suggesting the algorithm is suppressing consumer attention to certain hotels more than it is suppressing the quality of matches conditional on that attention. The DiM and covariate-adjusted estimates agree in direction and are stable across specifications, which rules out covariate imbalance as an explanation for the negative sign.

5.3 IV Estimates

Table 7 shows the first stage. Random ordering shifts a hotel’s displayed position by -0.39 slots on average (unadjusted), meaning hotels in randomly ordered searches tend to appear slightly higher on the page than they would under the algorithm. The first-stage F-statistics are 91.4 (unadjusted) and 102.6 (adjusted), far above the conventional weak-instrument threshold of 10. The instrument is strong and the relevance assumption is satisfied.

Table 7 reports the LATE estimates. Moving one position lower on the page reduces booking probability by 0.5 to 0.7 percentage points for complier hotels, and reduces click probability by 0.6 to 0.9 percentage points. These estimates should be interpreted carefully given the monotonicity concern discussed in section 6: they represent an average across hotels moving in both directions rather than a clean single-population effect.

6 Robustness and Limitations

Placebo outcome. If the randomization is valid, `random_bool` should not predict hotel price, which is fixed before the search occurs. The estimated effect on `price_usd` is -0.31 ($SE = 0.40$, $p = 0.43$). The null cannot be rejected. The randomization appears clean.

Subsample stability. Table 5 re-estimates the DiM ITT within six subgroups: domestic versus international searches, short versus long stays, and brand versus independent hotels. Every point estimate is negative and similar in magnitude to the full-sample result. Some subgroups lose statistical significance because they are smaller, not because the effect reverses. The result is not driven by any particular type of search or hotel.

Monotonicity. The algorithm is designed to promote higher-quality hotels: four- and five-star properties appear higher on the page under algorithmic ranking than under random ordering, while one- and two-star hotels are pushed lower. This is the algorithm doing exactly what it is supposed to do. However, it creates a problem for the IV design: the instrument moves high-quality hotels upward and low-quality hotels downward, so different hotels experience position changes in opposite directions. Monotonicity requires that the instrument pushes all hotels the same way, which it does not. As a result, the LATE conflates the effect of moving up the page for a high-starred hotel with the effect of moving down for a low-starred one, and cannot be cleanly interpreted as the effect for a single complier group. This is why the ITT remains the preferred estimand throughout. Future work could stratify the IV analysis by star rating tier to recover separate LATEs for hotels moving upward versus downward.

A second limitation concerns within-search interference. Displacing one hotel affects the visibility of its competitors on the same page. Applying the exposure mapping framework of Aronow and Samii (2017) to model these spillovers within search results is a natural extension of this analysis.

7 Conclusion

Using a randomized experiment embedded in Expedia’s search platform, this paper estimates the causal effect of algorithmic hotel ranking on bookings and clicks. The central finding is that algorithmic ranking reduces both outcomes relative to random ordering. The ITT estimates imply roughly a 9% relative reduction in booking probability, stable across specifications and subgroups.

The core result raises a simple but important question: who benefits from algorithmic ranking? The evidence suggests the algorithm primarily redistributes attention from lower-quality to higher-quality hotels. Whether this serves consumers who want better matches, or the platform whose revenue depends on bookings, or independent hotels displaced by branded ones, cannot be resolved from these estimates alone. But the finding that random ordering outperforms the algorithm on average is a useful starting point for that conversation.

References

- Aronow, P. M. and Samii, C. (2017). Estimating average causal effects under interference between units. *Annals of Applied Statistics*, 11(4):1912–1947.
- Expedia (2013). Personalize Expedia hotel searches – ICDM 2013.
- Ursu, R. M. (2018). The power of rankings: Quantifying the effect of rankings on online consumer search and purchase decisions. *Marketing Science*, 37(4):530–552.

Appendix

Table 1: Covariate Balance by Ranking Condition

Variable	Mean (Algo)	Mean (Random)	SMD
price_usd	147.81	148.12	-0.003
prop_starrating	3.296	3.211	0.097
prop_review_score	3.840	3.760	0.080
prop_brand_bool	0.663	0.675	-0.025
prop_location_score1	2.885	2.703	0.123*
promotion_flag	0.228	0.178	0.123*
srch_length_of_stay	2.113	2.057	0.035
srch_booking_window	31.590	30.280	0.029
srch_adults_count	1.941	1.951	-0.012
srch_children_count	0.341	0.345	-0.005
srch_saturday_night_bool	0.508	0.505	0.006
visitor_hist_adr_usd	11.324	11.674	-0.007
no_purchase_history	0.936	0.934	0.008
orig_destination_distance	1019.3	969.9	0.029
dist_missing	0.314	0.306	0.018
Searches	47,064	2,936	
(Search, hotel) pairs	1,153,157	68,683	
Booking rate	4.08%	4.27%	
Click rate	4.49%	4.72%	

Note: SMD = standardized mean difference. * indicates $|SMD| > 0.10$; both variables are included as controls in adjusted specifications. Randomization was conducted at the search level; mild hotel-level imbalance is expected due to within-search correlation.

Table 2: ITT: Effect of Algorithmic Ranking on Booking and Click Probability

	(1) Booking	(2) Booking	(3) Click	(4) Click
Algorithm ($Z = 1$)	-0.002** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	-0.004*** (0.001)
Controls	No	Yes	No	Yes
N		967.224		967.224
R ²	0.000	0.013	0.000	0.013

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

HC2 heteroskedasticity-robust standard errors in parentheses. Unit of observation is (search, hotel) pair. $Z = 1$ if search received algorithmic ranking, $Z = 0$ if random. Controls include hotel characteristics (star rating, review score, brand, location scores, price) and search characteristics (length of stay, booking window, party size, Saturday night, visitor history).

Table 3: First Stage: Effect of Random Ranking on Hotel Position

	(1) Unadjusted	(2) With controls
Random ranking (instrument)	-0.393*** (0.041)	-0.474*** (0.047)
Controls	No	Yes
First-stage F	91.4	102.6
N		967.224

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

HC2 robust SE. Dependent variable is displayed position (1 = top). A negative coefficient means random ranking pushes hotels higher on the page on average.

Table 4: LATE: Causal Effect of Position on Booking and Click Probability

	(1) Booking	(2) Booking	(3) Click	(4) Click
Position (LATE)	−0.005** (0.002)	−0.007*** (0.002)	−0.006*** (0.002)	−0.009*** (0.002)
Controls	No	Yes	No	Yes
First-stage F	91.4	102.6	91.4	102.6
N		967.224		967.224

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

HC2 robust SE. Instrument: random_bool. Position is the hotel's rank on the results page (1 = top). A negative LATE means being shown lower on the page reduces booking/click probability. Controls same as Table 2.

Table 5: ITT Stability Across Subsamples (Outcome: booking_bool)

Subsample	N	Estimate	SE	p-value
Full sample	1,221,840	−0.00193	0.00079	0.015
Domestic searches	797,950	−0.00145	0.00096	0.132
International searches	423,890	−0.00299	0.00141	0.034
Short stay (≤ 3 nights)	1,052,750	−0.00183	0.00085	0.032
Long stay (> 3 nights)	169,090	−0.00257	0.00215	0.233
Brand hotels	811,114	−0.00176	0.00097	0.069
Independent hotels	410,726	−0.00225	0.00138	0.105

Note: DiM estimator with HC2 robust standard errors. All point estimates are negative. Losses of significance in subgroups reflect smaller sample sizes, not directional inconsistency.

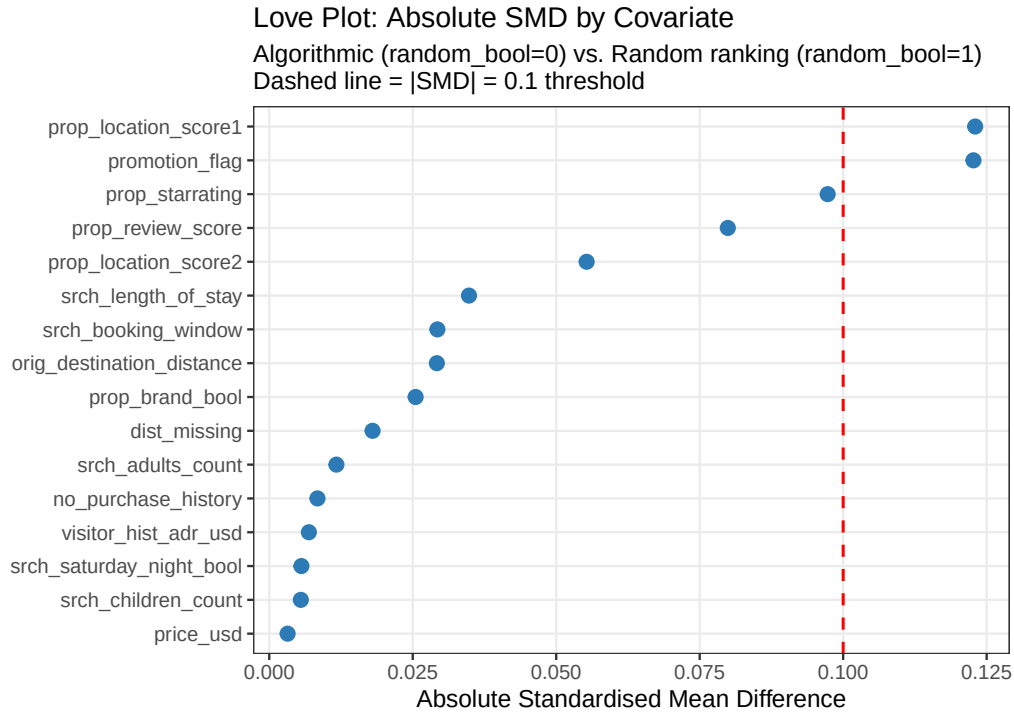


Figure 1: Love Plot: Absolute Standardized Mean Differences by Covariate. The dashed line marks the $|SMD| = 0.10$ threshold. Fourteen of 15 covariates fall below it, consistent with valid randomization of random_bool at the search level.

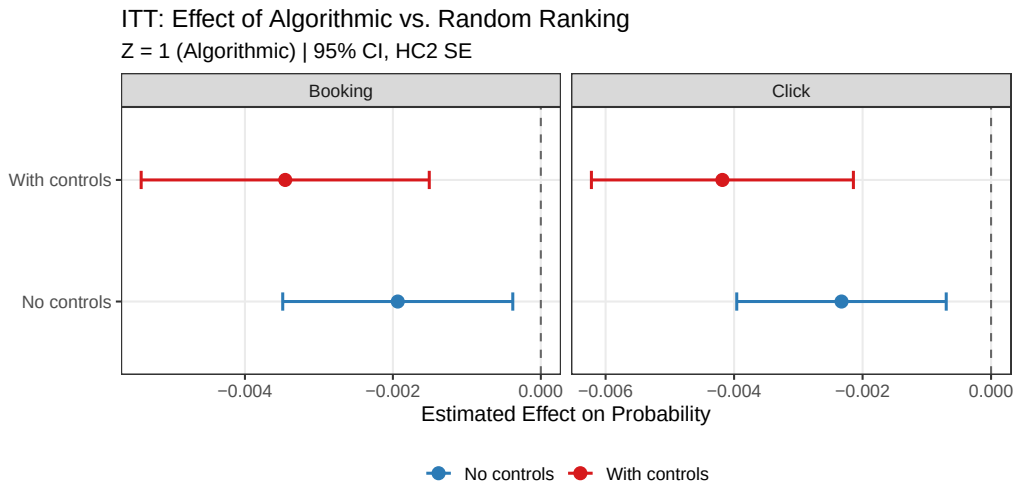


Figure 2: ITT Estimates with 95% Confidence Intervals. Left panel: booking_bool. Right panel: click_bool. Blue: unadjusted DiM. Red: covariate-adjusted (X_i included). All estimates are negative: algorithmic ranking lowers both booking and click rates relative to random ordering.